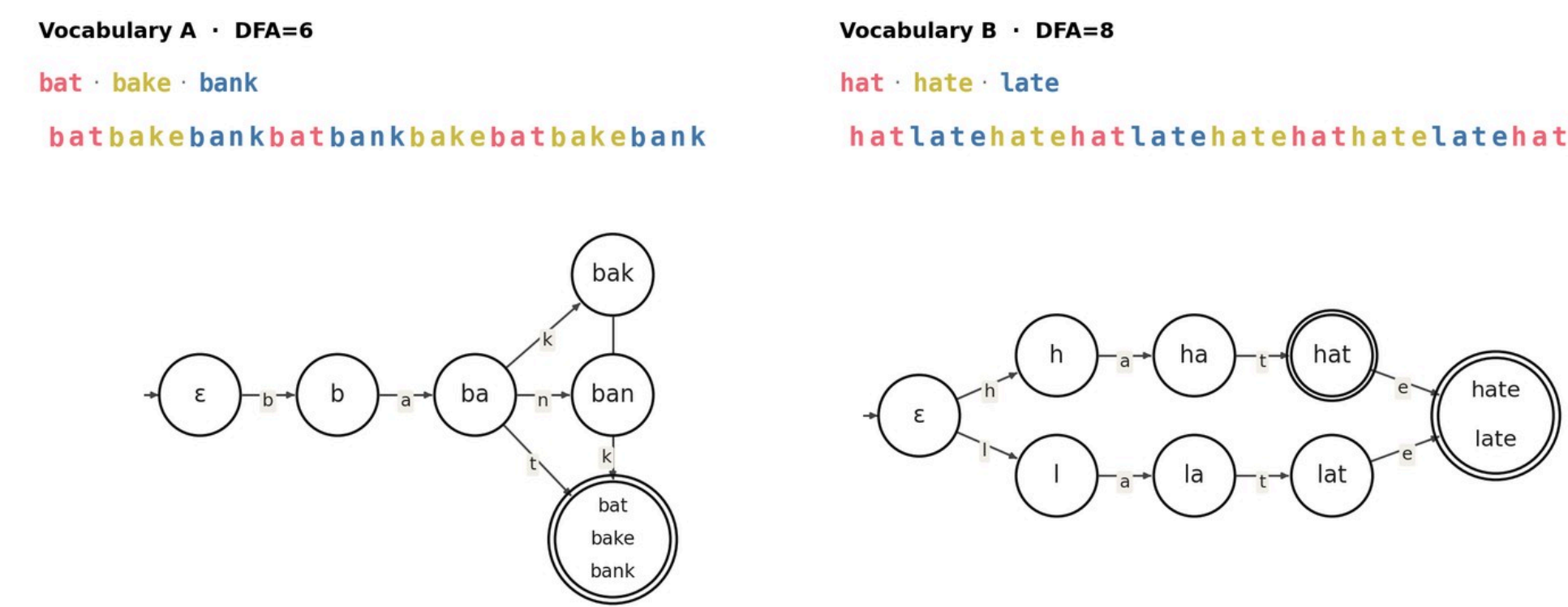


Aims

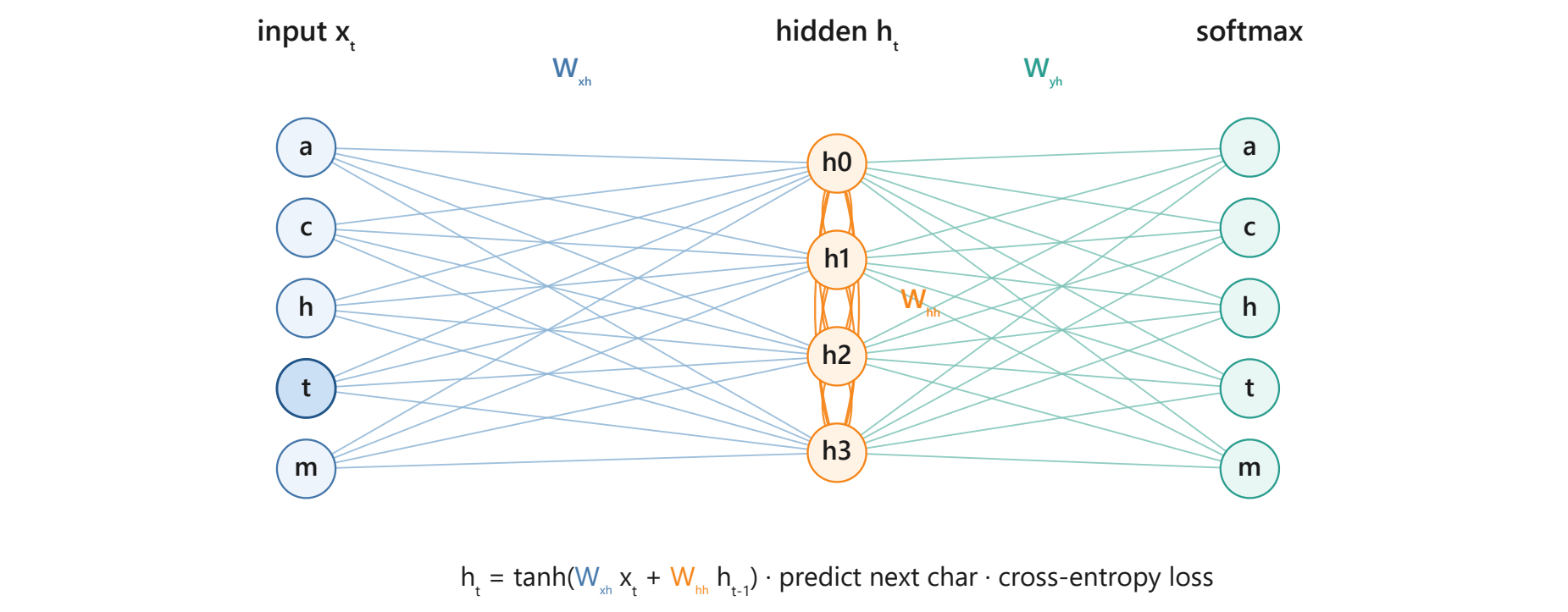
- How is lexical information (word position, character, etc.) represented in a recurrent network that learns to produce words?
- How does the network's representation of information change with vocabulary complexity?
- How do the network's weights differ as a function of the complexity of the learned vocabulary (function → structure)?

Fig 1 · Mixed vocabularies & minimal DFAs



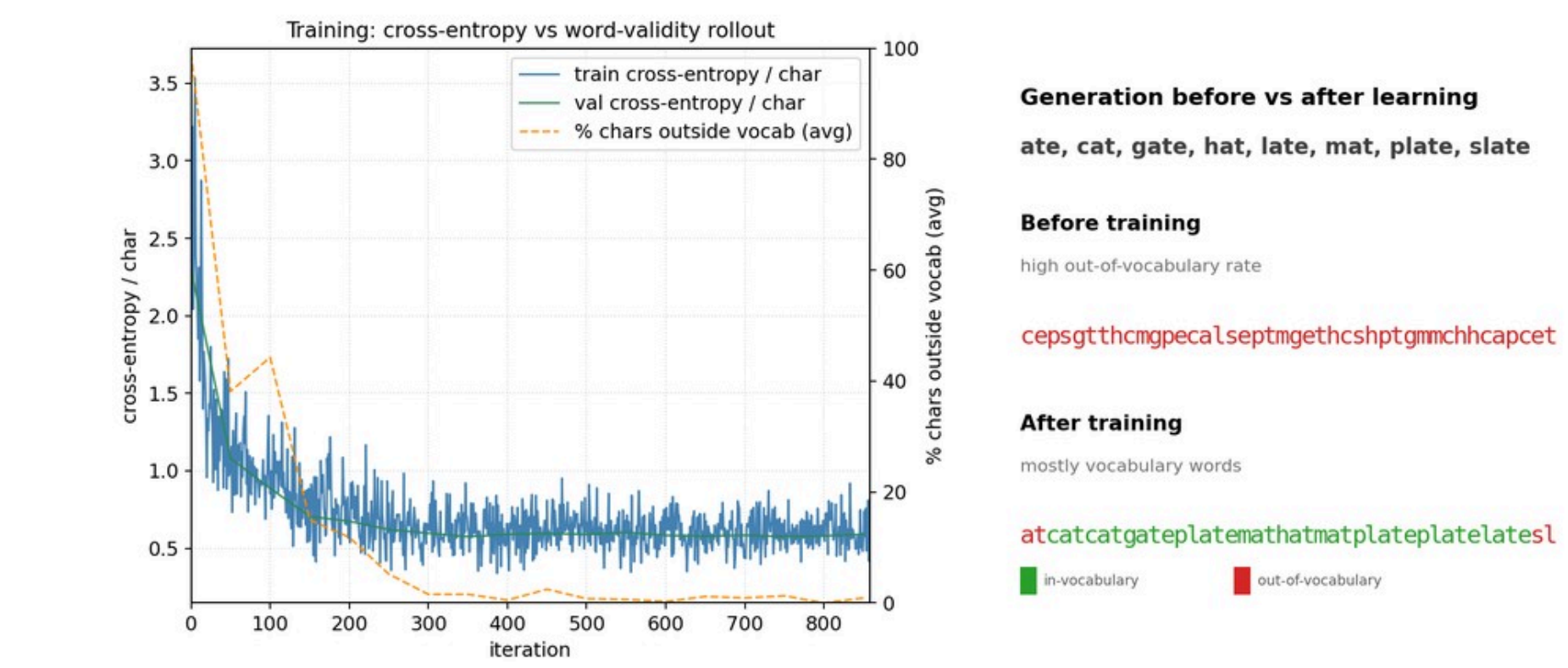
Two ≤3-word conditions. Vocabulary words (bold), one unsegmented stream, then the prefix-labeled minimal DFA (6 vs 8 states).

Fig 2 · Training the RNN



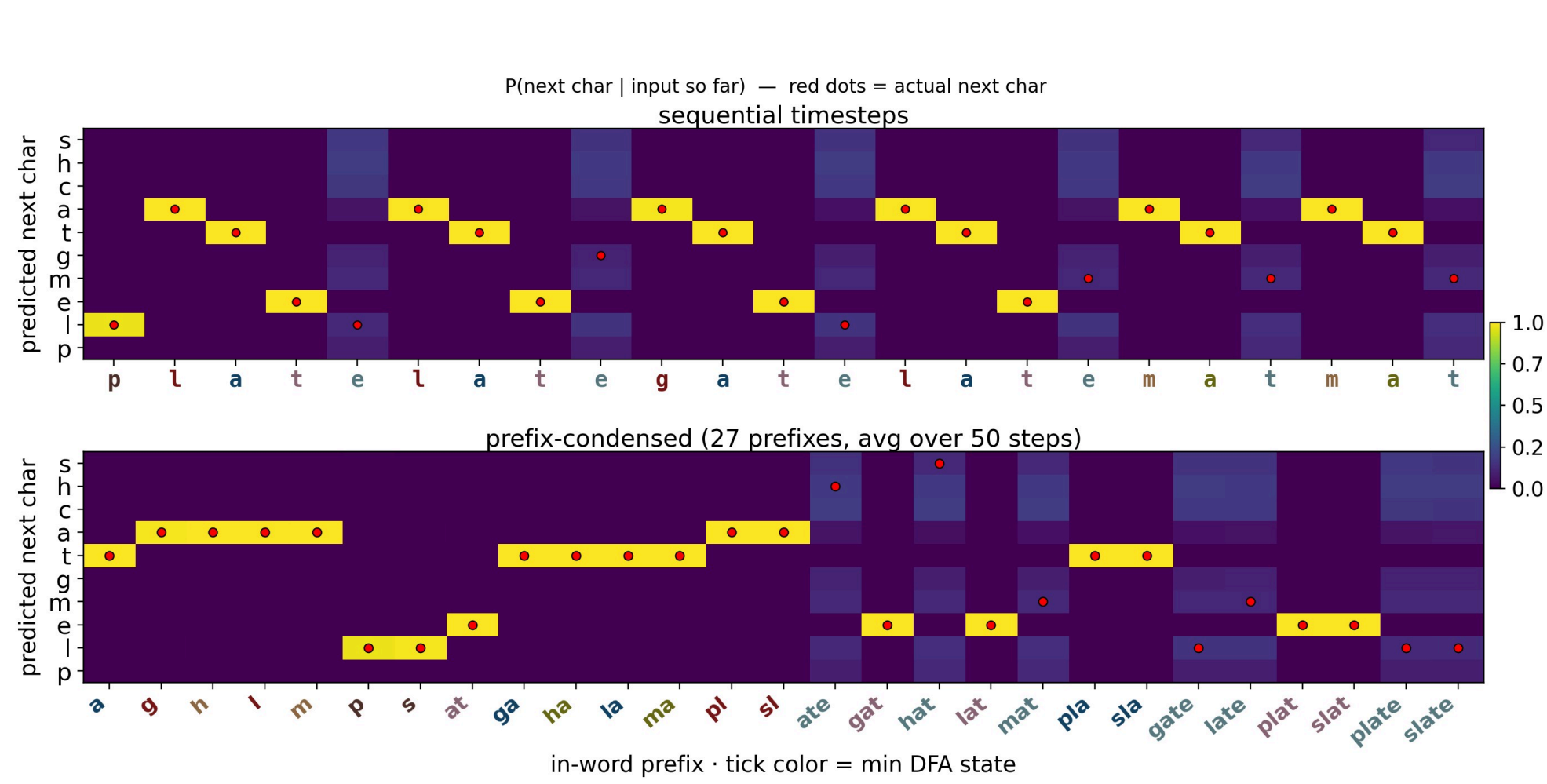
Character-level Elman RNN. Input x_t predicts the next character via $h_t = \tanh(W_{in} x_t + W_{hh} h_{t-1})$; cross-entropy loss.

Fig 3 · Learning & generation



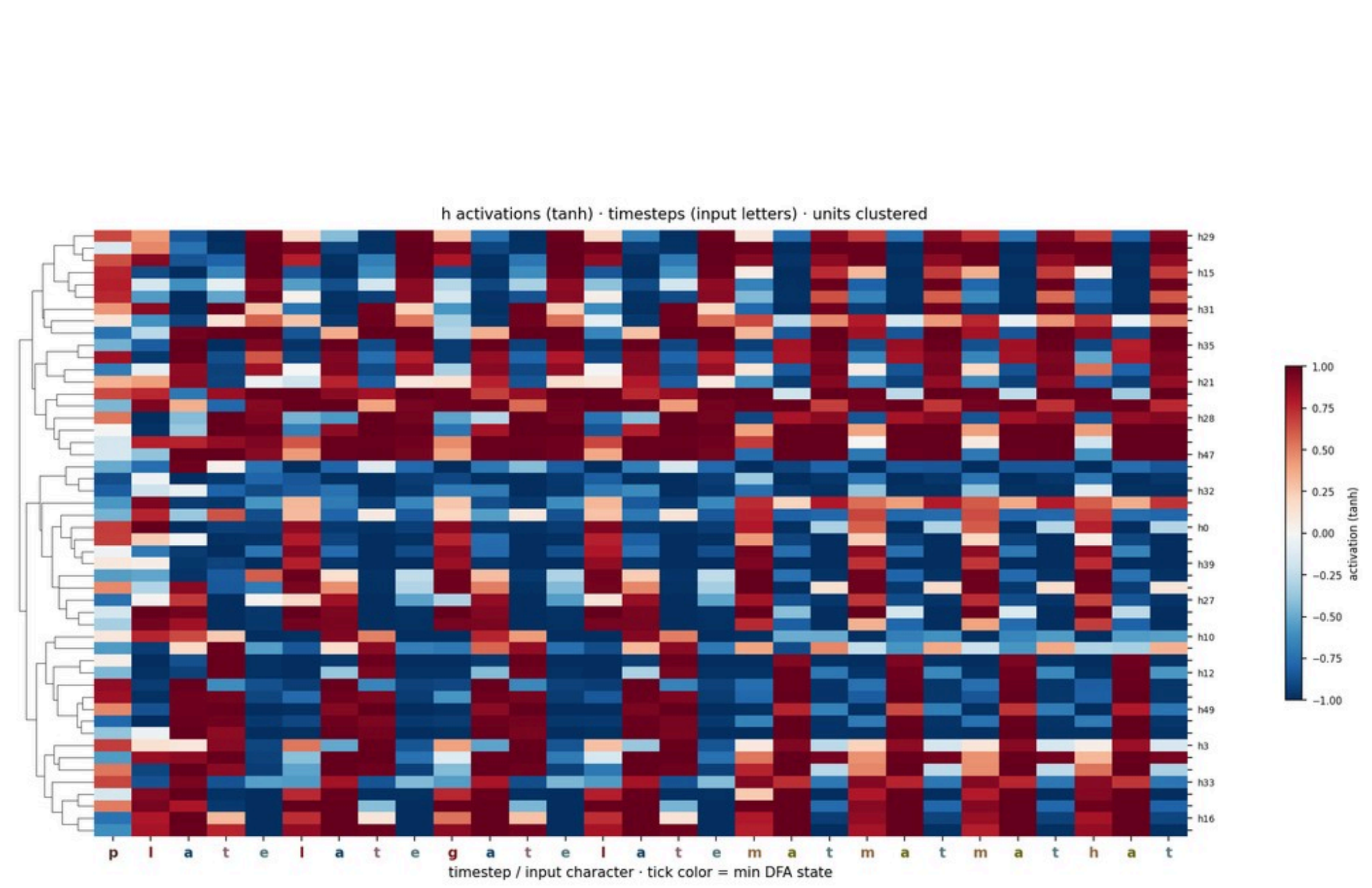
Loss and OOV collapse together; rollouts become legal vocabulary words. Demo lexicon listed above the panels.

Fig 4 · Next-character probabilities



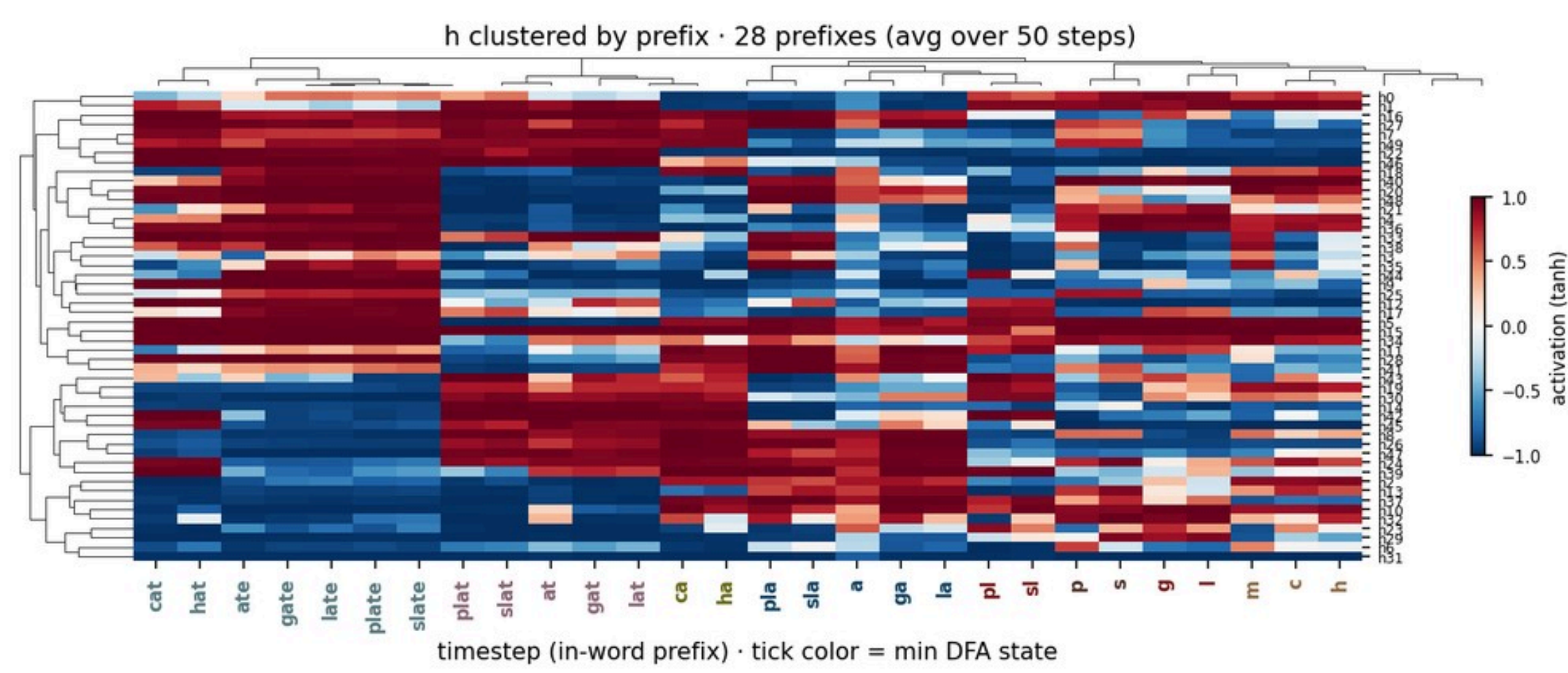
Softmax $P(\text{next char} | \text{input so far})$. Red dots = actual next character.

Fig 5 · Activation heatmap



Hidden units over timesteps (input letters; units clustered).

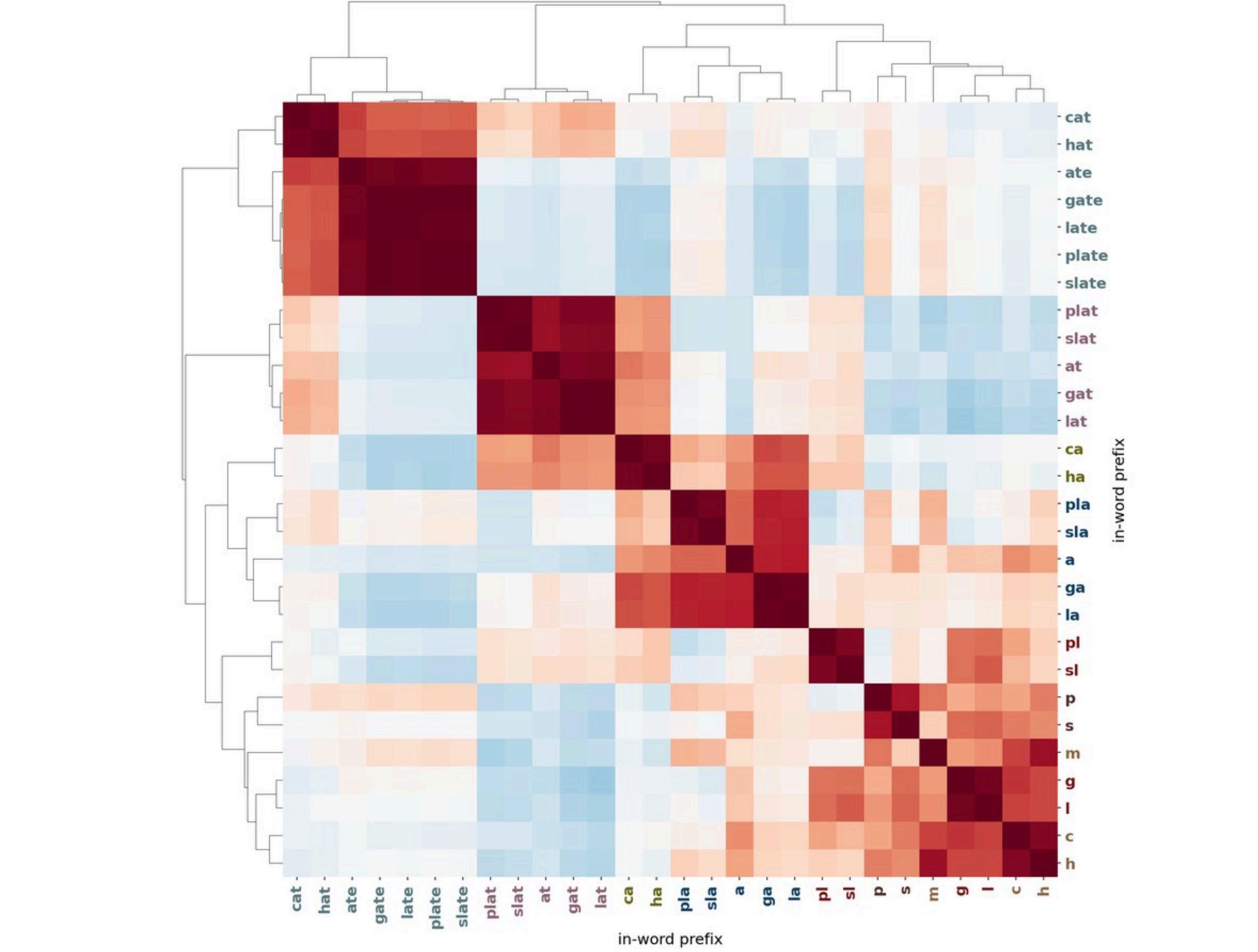
Fig 6 · Activations by prefix



Activations clustered by in-word prefix. Tick color = DFA state.

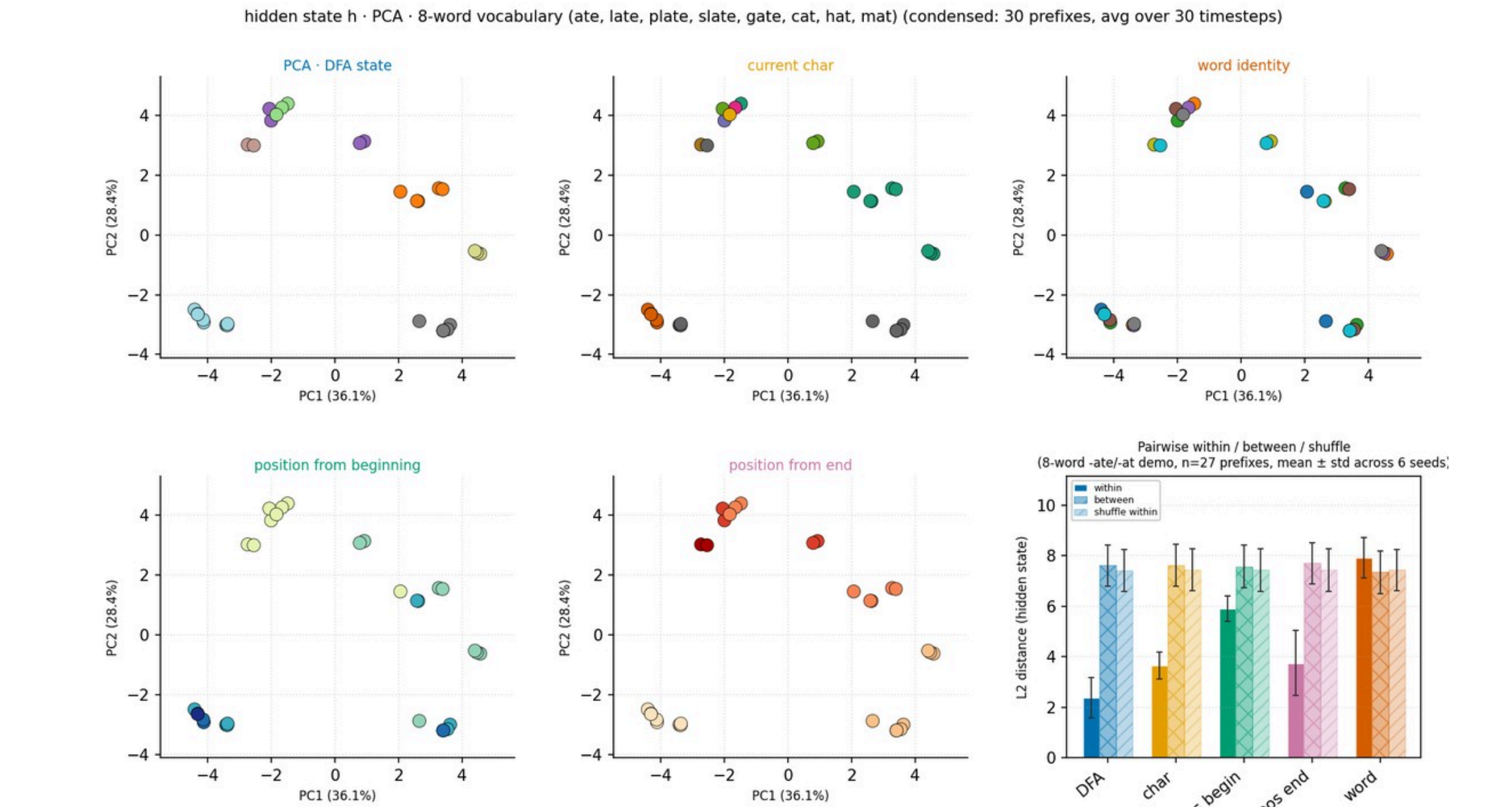
Fig 7 · Hidden-state correlation

hidden state correlation · tick color = min DFA state · vocabulary: ate, late, plate, slate, gate, cat, hat, mat · chars: shamcpetg (condensed: 28 prefixes, avg over 50 timesteps)



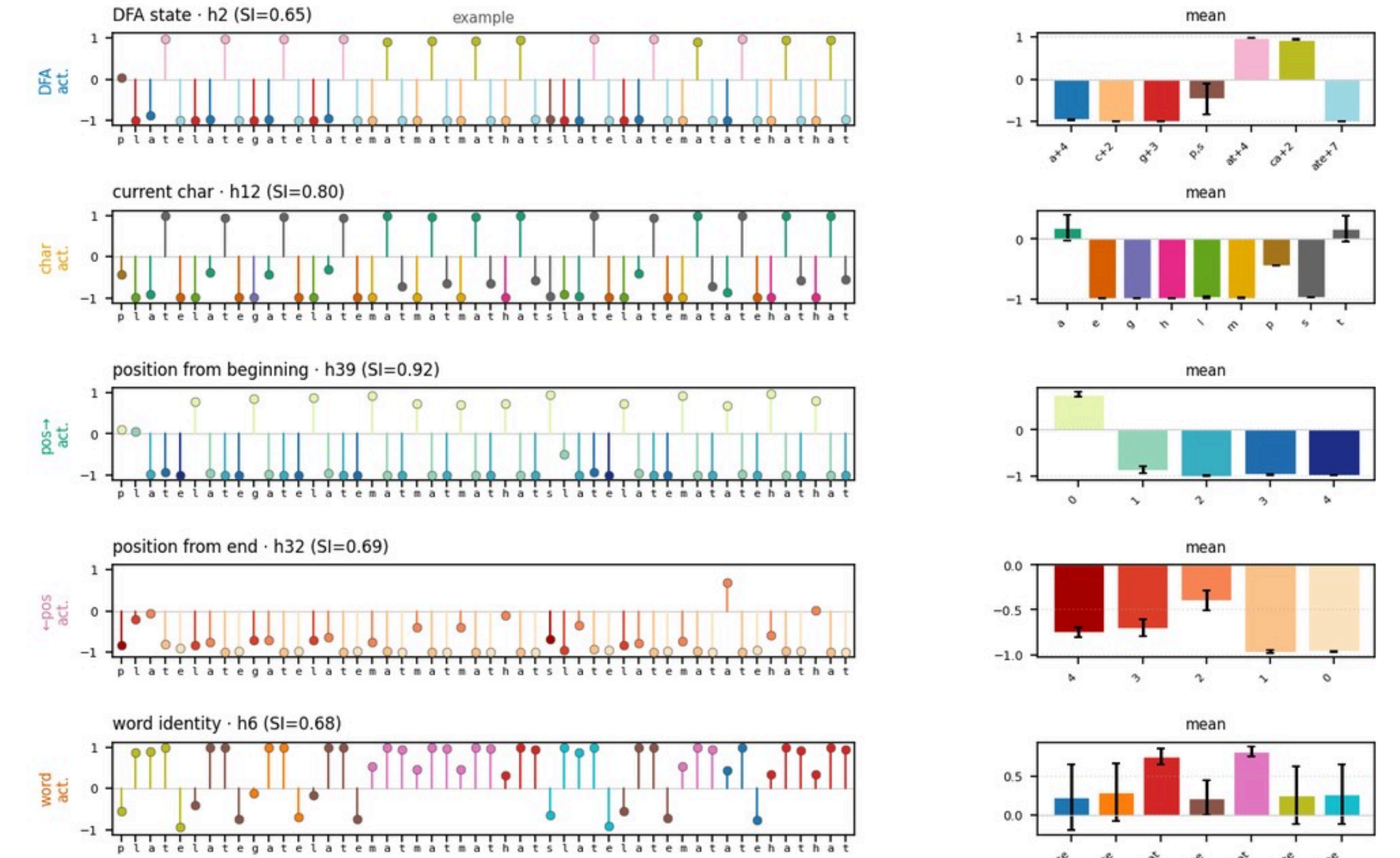
Prefix × prefix correlation. Blocks merge prefixes that share a future.

Fig 8 · DFA geometry & separation



PCA of h_t by feature; pairwise within/between/shuffle at end of row. DFA $\eta^2 \approx 0.88$; word identity ≈ 0.06 .

Fig 9 · Example selective units



One top unit per feature on a shared corpus window.

Fig 10 · Linear decoding

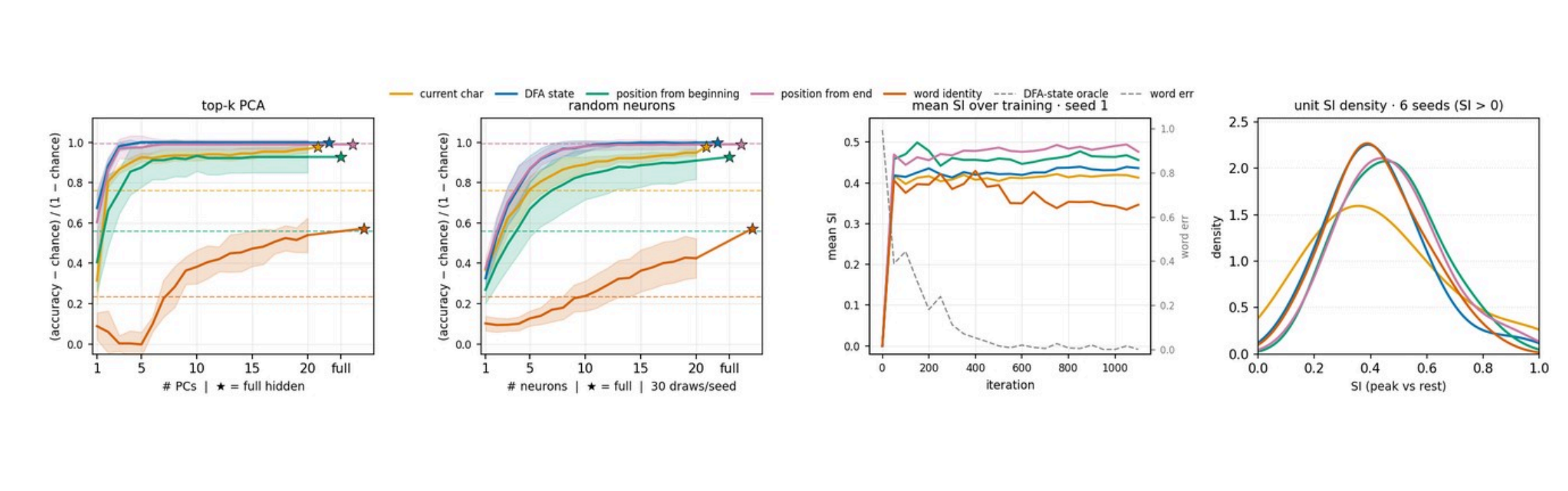
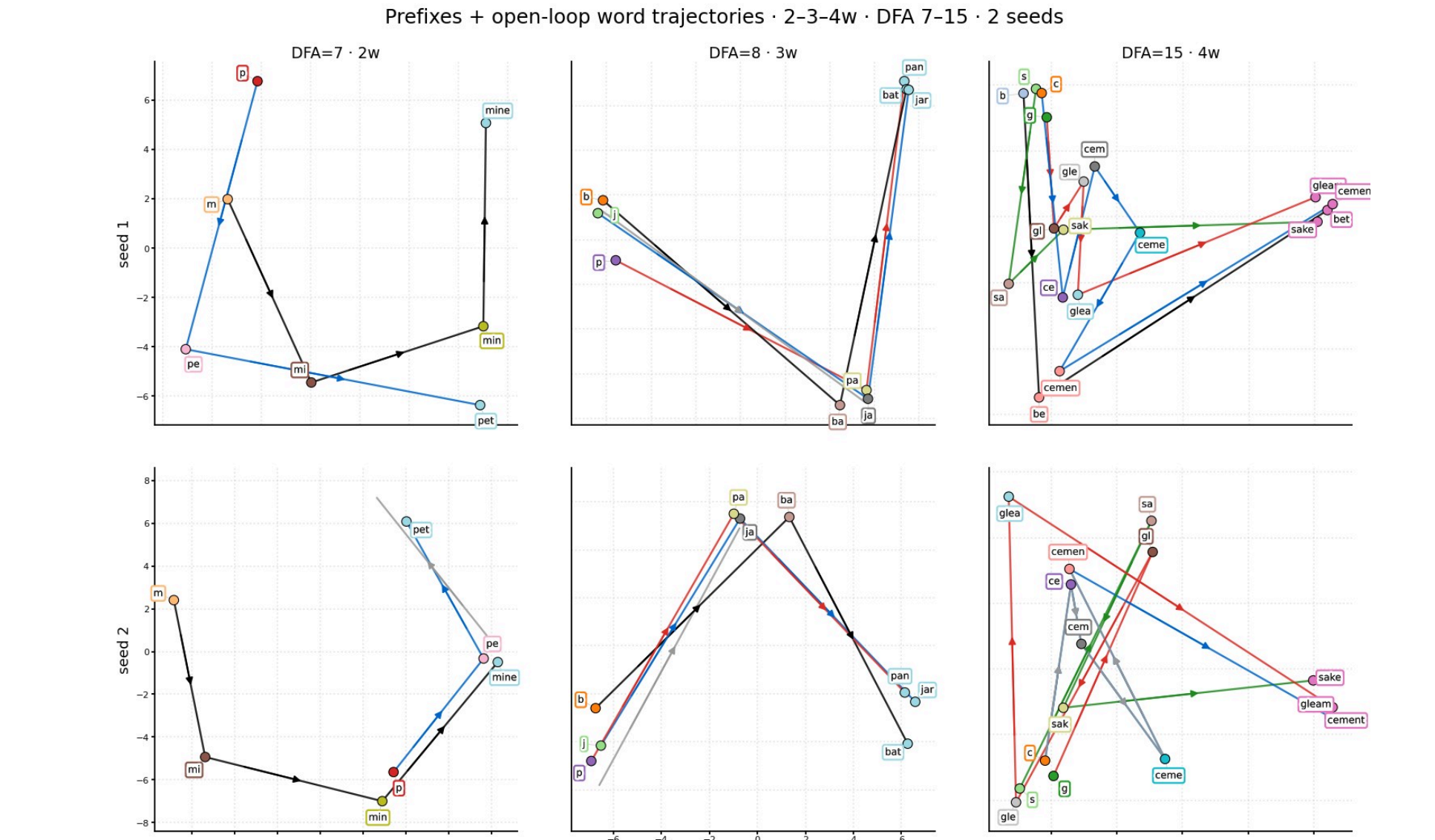
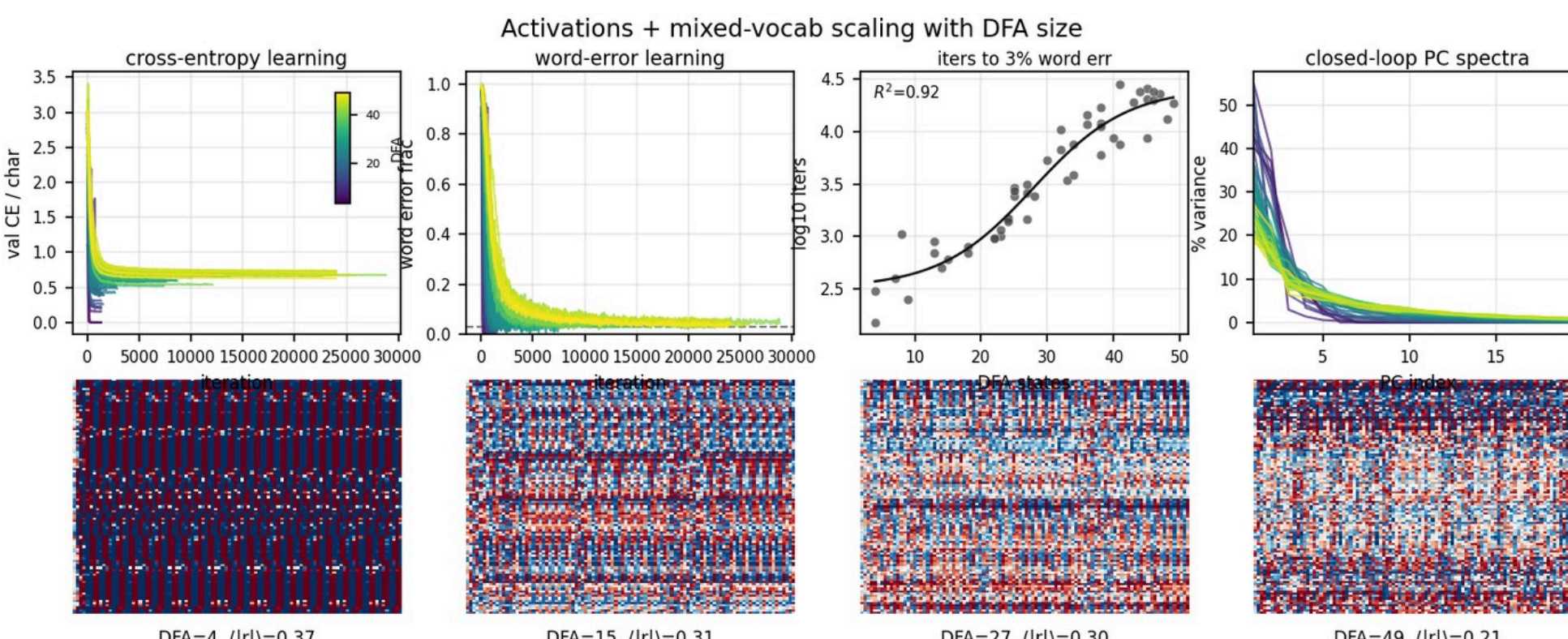


Fig 11 · Word trajectories



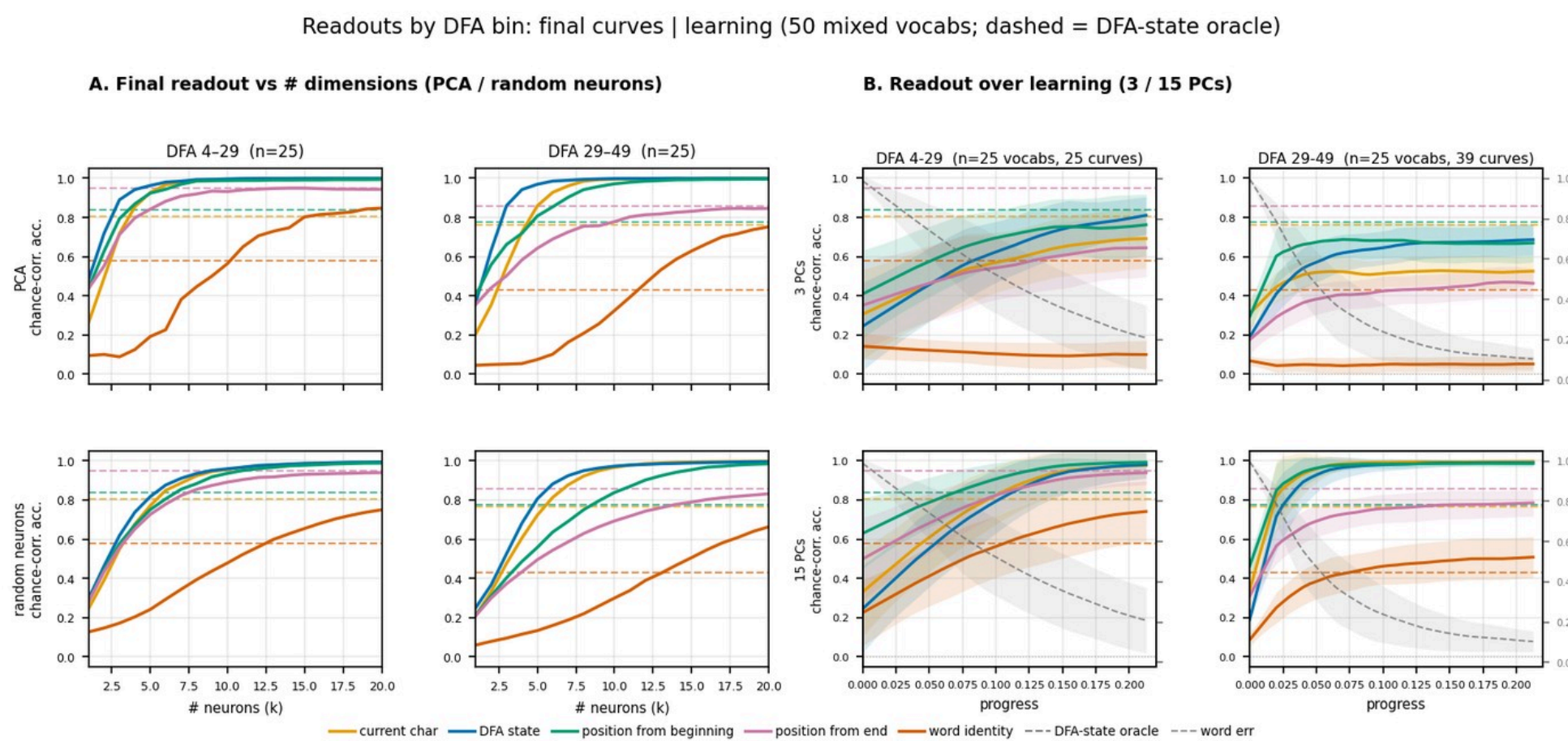
DFA-labeled prefixes with open-loop word trajectories overlaid (same PCA plane). Columns: 2-4 word mixed vocabs; rows: two seeds.

Fig 12 · Activations & scaling vs DFA size



Top: four activation rasters spanning DFA size. Bottom: CE and word-error learning, iters to 3% WE, and closed-loop PC spectra.

Fig 15 · Readouts vs DFA size



Two DFA bins. Left: final PCA/heuron readout. Right: readout over learning (3/15 PCs). Word identity needs ~15 PCs.

Fig 16 · Geometry & unit selectivity

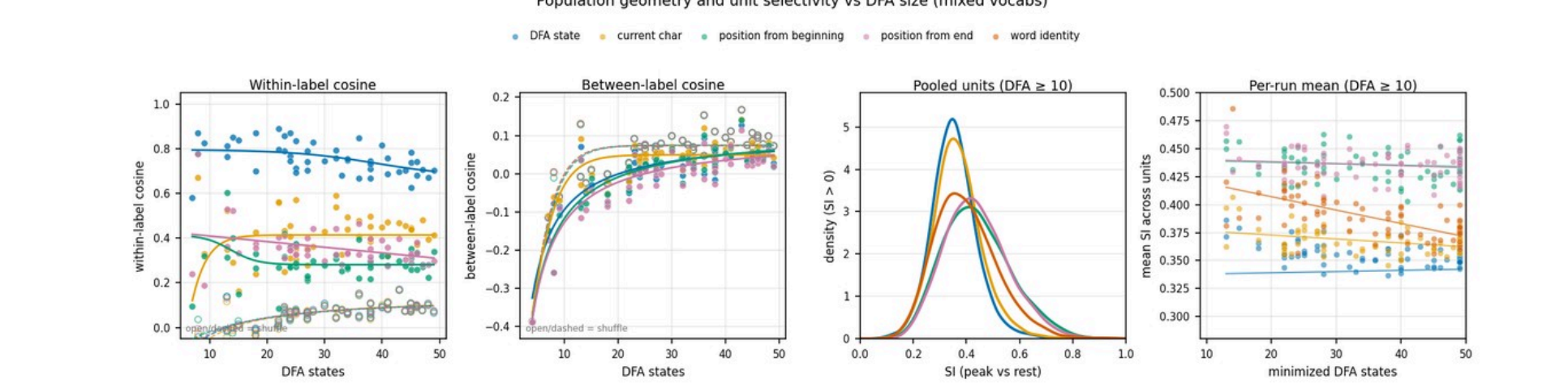
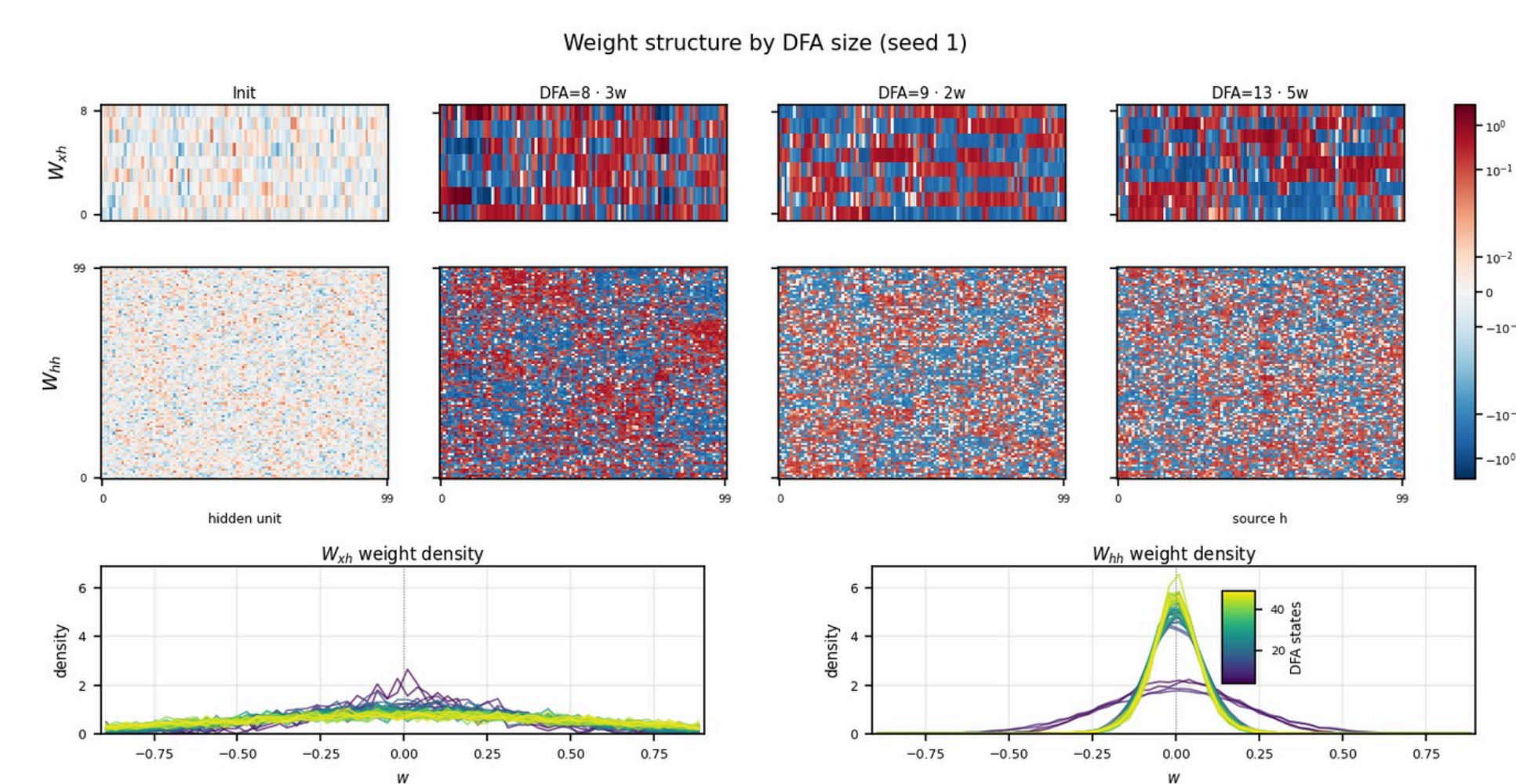
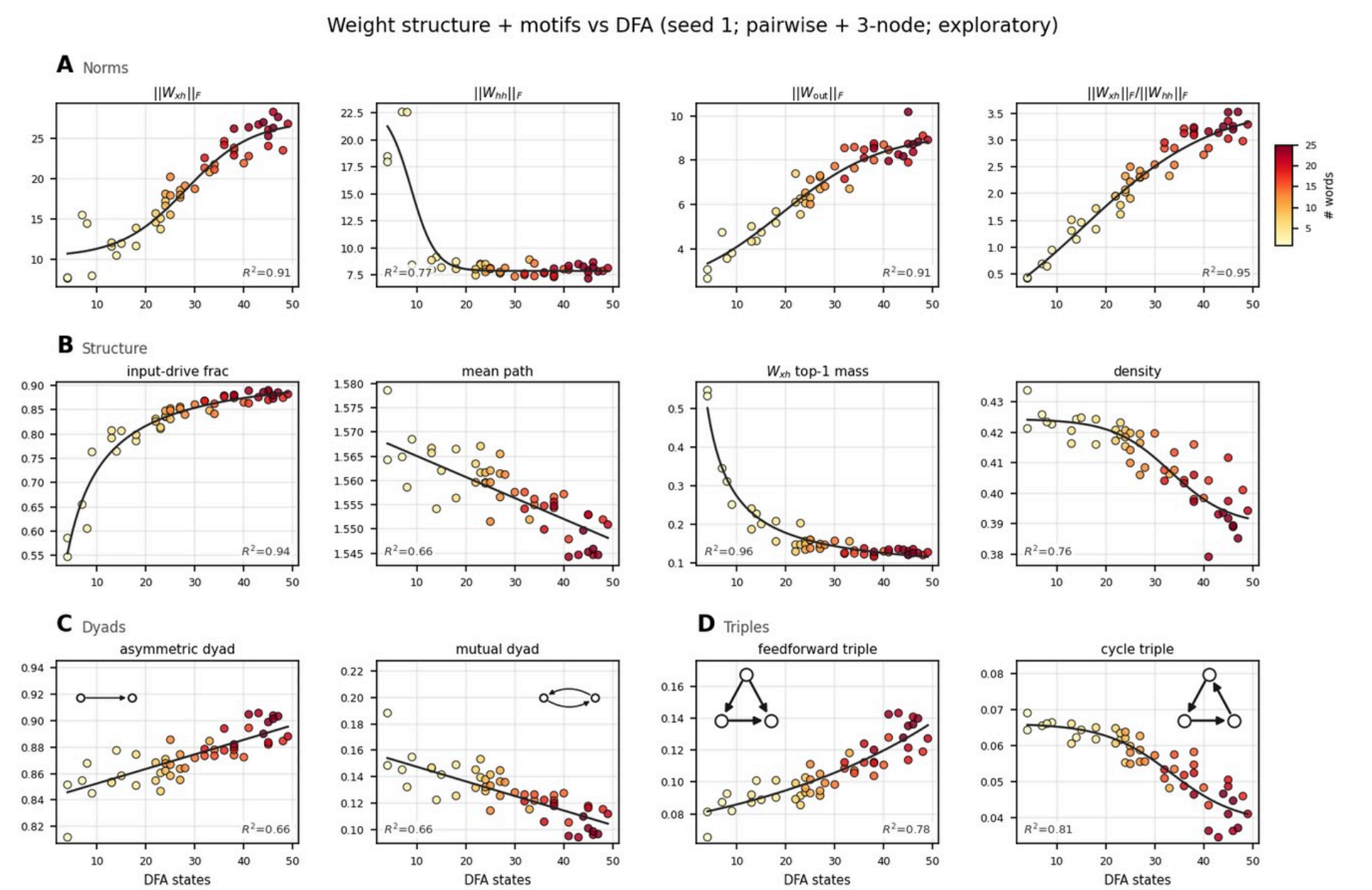


Fig 18 · Weight matrices vs DFA



Init + DFA 8/9/13 matrices (shared symlog w scale); bottom: W_{in} / W_{hh} densities (color = DFA size).

Fig 19 · Weight graph motifs vs DFA



Norms, structure, pairwise and 3-node motifs. Larger DFAs shift feedforward.